

2026 AZA CONSERVATION GRANTS FUND

Cover Sheet

Title of Project: Assessment and Conservation of Nebraska's Only Nocturnal Tree Glider: The Southern Flying Squirrel

Student Name: [REDACTED]

Institution: University of Nebraska at Omaha

Project Duration: January 2026 – December 2026

Project Category: Research and Animal/Population Management

Project Abstract

Within Nebraska, the Southern Flying Squirrel (*Glaucomys volans*) historically inhabited the southeastern portion of the state. This area today is known as Douglas, Sarpy, Otoe, Nemaha, and Richardson counties, all of which border the Missouri River. These species thrive in ecosystems rich in mature oak and hickory forests. However, due to urbanization, the conversion of land to agricultural use, and suppression of fire, mature oak savannas and oak woodlands are recognized as one of the leading endangered natural communities worldwide

Lacking and improperly managed ecosystems have degraded the habitat availability for the Southern Flying Squirrel in Nebraska. Due to this, their numbers have dwindled and are now recognized as a threatened species in the state of Nebraska. The only known population of SFS is in Indian Caves State Park, and a newly discovered population in 2018 in Lincoln, NE. Although SFS are known to be inhabiting ICSP, there is no current count of individuals that make up this population, or any known information on how these species are living in a semi-restored, semi-degraded environment. Without the restoration of this population, no SFS will live in native habitats throughout Nebraska.

The scope of this project plans on documenting and researching the ICSP population of SFS for improved habitat of oak woodland management, assessing viability for potential translocations or reintroductions, and generating a population baseline for Nebraska Game & Parks. In addition, partnerships with AZA zoos for ex situ programs such as breeding, reintroductions, and future assurance populations.

Project Narrative

Project Overview & Significance

Historically, Nebraska's native landscapes encompassed an elegant mosaic of oak woodlands, oak savannas, and prairie ecosystems. However, due to habitat fragmentation and degradation, these ecosystems are declining. In Nebraska, along the Missouri River, there are far and few areas that are preserved and managed for these natural landscapes. It is of high importance that restoration is prioritized for the health of these environments and the species that encompass them. Understanding the way in which the Southern Flying Squirrel interacts within its native habitat can guide land management in future restoration projects and goals. In addition, habitat improvement can improve the threatened status of this species with adequate conservation actions.

Goals, Objectives, & Actions

Project Goal: Over the course of one year, this project's goal is to successfully identify, tag, and genetically sample the population of Southern Flying Squirrels in Indian Caves State Park. Allowing for further understanding of what management steps need to be taken in the future, such as reintroduction programs, old-growth forest protection, and deciding on canopy tree species abundance. If reintroduction were a future step, goals would include a population that is self-sustaining and genetically diverse, and would create a model for other parks and preserves in the native SFS habitat range to follow. These goals align with the AZA *One Plan Approach* for in situ conservation.

Short-term goals consist of understanding population data to be used as a baseline for Nebraska Game and Parks, and for genetic use for understanding where the newly found Lincoln, Nebraska population originated from. In cohesion, the long-term goals consist of continued and improved management of oak woodland parks and preserves in Douglas, Sarpy, Otoe, Nemaha, and Richardson counties, with the intent of creating a climate change-resilient and healthy ecosystem in which native flora and fauna can thrive. Creating habitats where the introduction of this glider species is possible can be a step towards paving the way for other lost or dwindling fauna, such as bison.

Project Objective: Assess the current population size of the Southern Flying Squirrels in Indian Cave State Park to understand the behavior and health of individuals, provide information for management of oak woodlands throughout Eastern Nebraska, and understand future management needs for the park and other managed areas.

Actions: To assess the current population size of the SFS in ICSP, actions such as camera and live trapping, tagging, acoustic monitoring, box nesting sites, and DNA sampling will occur throughout the one-year time frame. These actions will increase knowledge of population size, health, habitat preference, and stability. The gathered data will be analyzed and written into a scientific peer-reviewed article that can be used by other preserves and parks to understand how they can implement certain management practices to move towards the reintroduction of this species. Management practices will be based upon gathered information of SFS habitat, health, and prominence in ICSP. Post project, a next-step plan will be created for ICSP for this Nebraska Native threatened species. Depending on observed data, this could consist of reintroduction, relocation, or improved management in particular areas.

Methods

SFS are a nocturnal, secretive species that are not often seen in the wild. So, to be able to document the number of individuals within the ICSP population, varying trapping and observational methods need to be used. This will include setting up trail cameras, live traps, and implementing nest boxes. 20 trail cameras will be placed throughout the 3,400 acres of property (about 15-20ft in height), spacing them out to have 10 in restored and managed areas, 5 in untouched degraded areas lacking in old growth and dead trees, and 5 in areas close to and along trails. Cameras will be placed on varying tree species such as oak varieties, hickories, and basswood. Footage will be used to identify habitat ranges of SFS and to document how many individuals are seen. Bi-weekly live trapping outings will occur starting in April; three live traps will be set out in the trail camera locations. Traps will be baited and will be opened before dusk and closed in the morning if no specimen is taken. These trappings will provide the opportunity to mark and release individuals, as well as take vitals to understand the health of the population. ICSP has a few pre-existing nest box sites (no information can be found on how many boxes there are or if any inhabitants are prominent); however, 5 new nest box sites will be implemented in restored, untouched, and public areas. These nests will give the ability to take DNA samples to understand genetic diversity between individuals and account for how many individuals are present. Lastly, during live trapping nights, nighttime acoustic monitoring will occur in separate areas to see if there is any activity in other portions of the park.

Timeline

Starting in January of 2026, supplies will be bought and prepared, and a trip of the project staff and volunteers will be taken to ICSP. Here, the park will be assessed, and trail camera, trapping, and nest box sites will be chosen and mapped out using the onX mapping tools. In February, a second trip will be taken to set up the trail cameras and nest box sites; however, live trapping will not begin until April, as that is when the SFS activity level increases. A third trip in March will occur to check on sites, change batteries if needed, and inspect nesting boxes. From April through October, bi-weekly (twice a month) site visits will take place to perform baited live trapping. These events will occur over a two-night period. In November, live trapping will be halted due to weather and the SFS being less active; however, a monthly check-in of equipment will still occur. This month, data will start to be compiled and assessed. In December, data collection activities will be halted, and a gathering will be held to showcase findings to the public.

Evaluation & Outcomes

Outcomes will be evaluated based on the goal of increasing understanding of the current Southern Flying Squirrel population in ICSP. If no species are identified, that information will be given to Nebraska Game & Parks and used to increase the species from threatened to endangered status. The population data received can provide an indication such as the health of the ecosystem, what areas of the park provide sufficient habitat, what areas do not, and plausible introduction sites.

Dissemination

As stated in the timeline section, results will be showcased at a public event held by ICSP in December. A presentation will be given by the project team, describing the findings. The event will also have guest speakers from the Henry Doorly Zoo, Nebraska Game & Parks, and Fontenelle Forest to describe the importance of zoos in the reintroduction of species, what the NGP's plan is with the newly found Lincoln population, and viewpoints from other preserves on what reintroduction programs might look like for them in the future. The event will be educational and get the public involved in what ICSP is doing to understand the SFS in Nebraska and why it is important for the future health of the ecosystem.

Ethical & Welfare Consideration

As this project actively involves being in close contact with the Southern Flying Squirrels, ensuring the appropriate care and welfare of these mammals is of great importance. During live trapping events, the traps will be opened before dusk and will be closed again in the morning. Traps will not be left open when the project team is absent to reduce the risk of the SFS and other animals from getting injured. Only those qualified on the project staff will be able to handle the mammals. No permits will need to be obtained for transportation since the SFS will be assessed on-site. For human protection, those handling specimens will be required to be up to date on all relevant vaccinations and will be required to wear gloves and masks.

BUDGET	COST ESTIMATION	DESCRIPTION
MOULTRIE EDGE CELLULAR TRAIL CAMERA – 1 PACK – X 20	\$1,000	Allows it to connect to the strongest signal from a variety of nationwide network so you don't have to swap SIM cards.
MOULTRIE EDGE MOBILE APP – PER MONTH – X 12	\$120	Monthly subscription to receive real time camera data updates
HAVAHART EASY SET SMALL 1-DOOR ANIMAL TRAP - X 5	\$230	Live traps
SONG METER MICRO 2 - X 2	\$300	Acoustic monitoring device
JCS WILDLIFE SQUIRREL HOUSE NESTING BOX – X 5	\$500	Mountable nesting boxes
BAITING INGREDIENTS	\$100	Bait for live traps
ON-X MAP - PER YEAR	\$100	Mapping subscription
HOUSEING – CAMPGROUND TENT SITE TRAVEL	\$500	Campground sites at ICSP
PROJECT MANAGER STIPEND	\$3,000	Gas & milage
ASSISTANT PROJECT MANAGER – X 2	\$3,000	Year stipend
VOLUNTEER UNDERGRADUATE STUDENTS	\$4,000	Year stipend
GENETIC SEQUENCING	\$0.00	Not paid
HANDLING GLOVES, TAGGING TOOLS, BAGS	\$3,000	Population Genetics data
EQUIPMENT REPAIR OR REPLACEMENT	\$500	Gear for tagging and handling SFS
EDUCATIONAL OUTREACH MATERIALS	\$300	For Repairs and Replacements
TOTAL	\$500	Rental space, guest speakers, event set up
	\$17,150	

Budget Justification

The AZA Conservation Grants Fund will fund the project's goals of research and animal/population management. No other funding has been granted towards this project.

A third of the funding will go to obtaining the needed equipment to monitor the SFS. This consists of trail cameras, live traps, bait, acoustic monitoring devices, and nesting boxes. This equipment is needed to adequately monitor and assess the full scope of the population. Another third of the project's budget goes to the project manager and assistant project manager's stipends. If this project is only partially funded, the stipends will be removed, and equipment will be prioritized. The last third of the project's costs goes to genetic sequencing of specimens, equipment for human and animal welfare and care, gas and mileage, and outreach materials for dissemination purposes

References

Daniel C. Dey, John M. Kabrick, Callie J. Schweitzer, Silviculture to Restore Oak Savannas and Woodlands, *Journal of Forestry*, Volume 115, Issue 3, 1 May 2017, Pages 202–211, <https://doi.org/10.5849/jof.15-152>

Dey, D. C.; Kabrick, J. M.; Schweitzer, C. J (2016). "Silviculture to restore oak savannas and woodlands". *Journal of Forestry*. 115 (3): 202–211. doi:10.5849/jof.15-152

Glaucomys volans (southern flying squirrel) | INFORMATION | Animal Diversity Web. (n.d.).

https://animaldiversity.org/accounts/Glaucomys_volans/#:~:text=Although%20they%20are%20not%20restricted%20to%20any,both%20local%20availability%20of%20cavities%20and%20climate.

Southern Flying Squirrel. (n.d.). https://www.esf.edu/aec/adks/mammals/southern_flying_squirrel.php